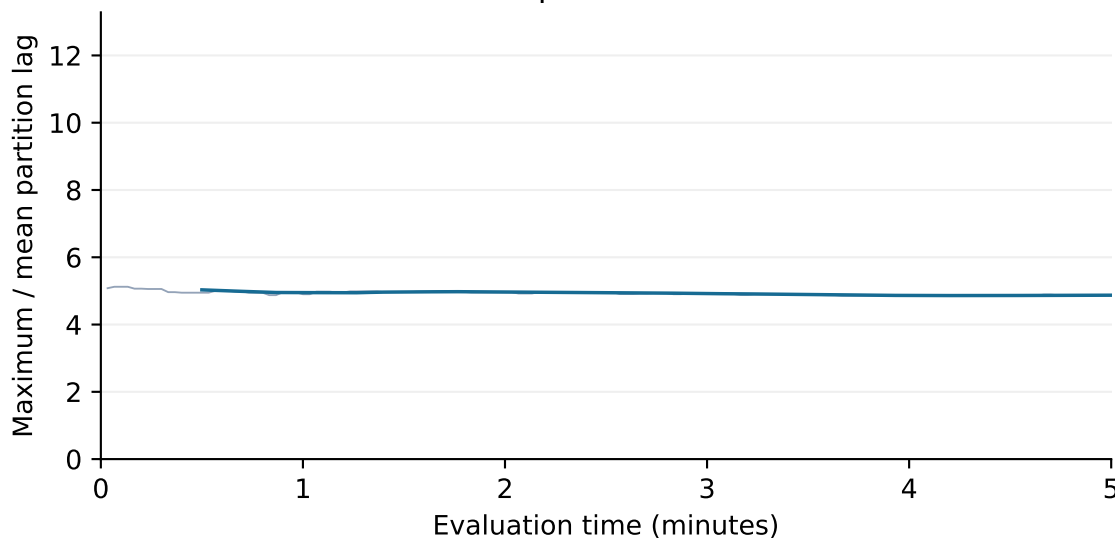


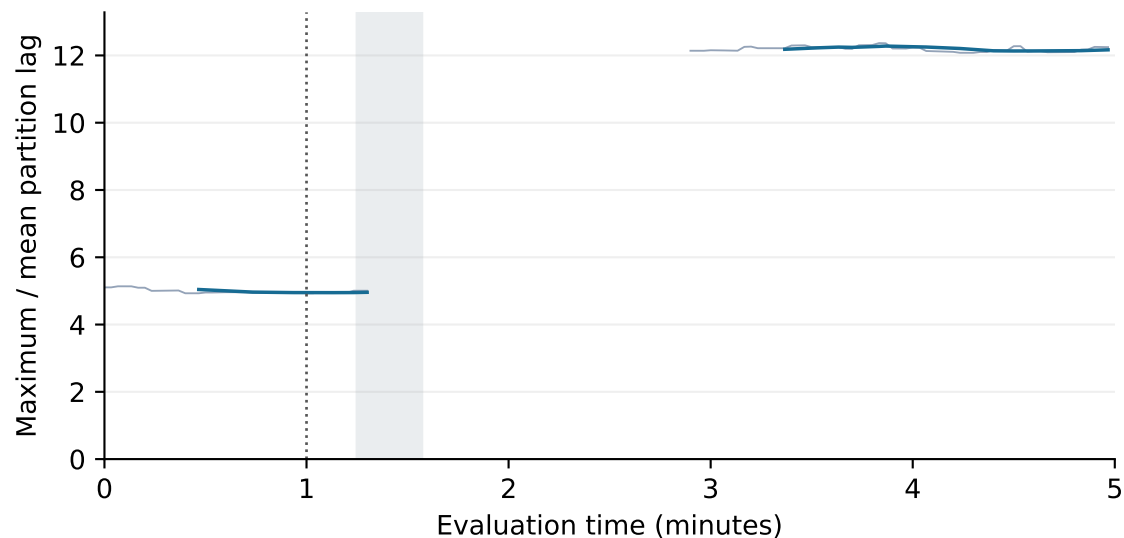
Partition lag skew — 700 messages/s

— Snapshot ratio — 15-sample rolling mean

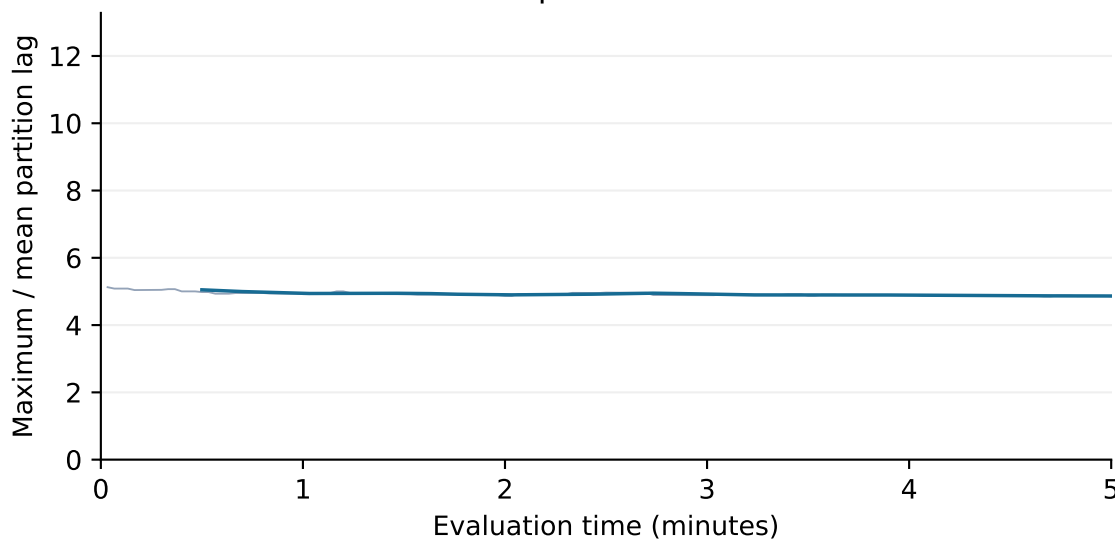
Keep 3 — Run 1



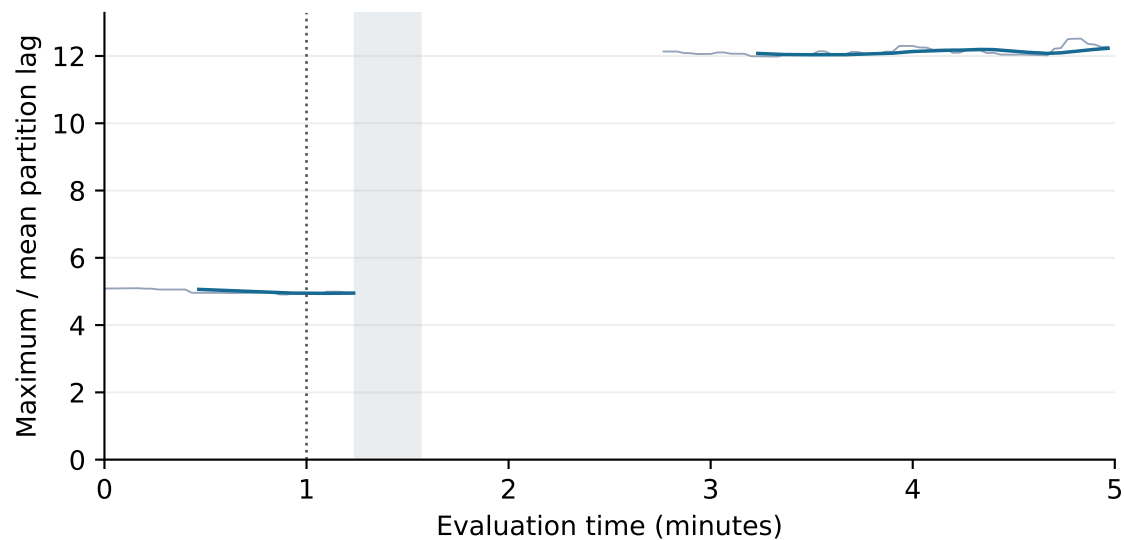
Redistribute within 3 — Run 1



Keep 3 — Run 2



Redistribute within 3 — Run 2



Common scales. Dotted line: scheduled redistribution request; shading: handoff coordination. Gaps remain unavailable. Interpret relative skew together with absolute lag and growth; a high ratio alone does not imply a large backlog.